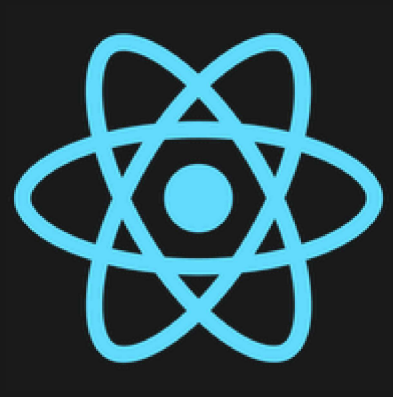
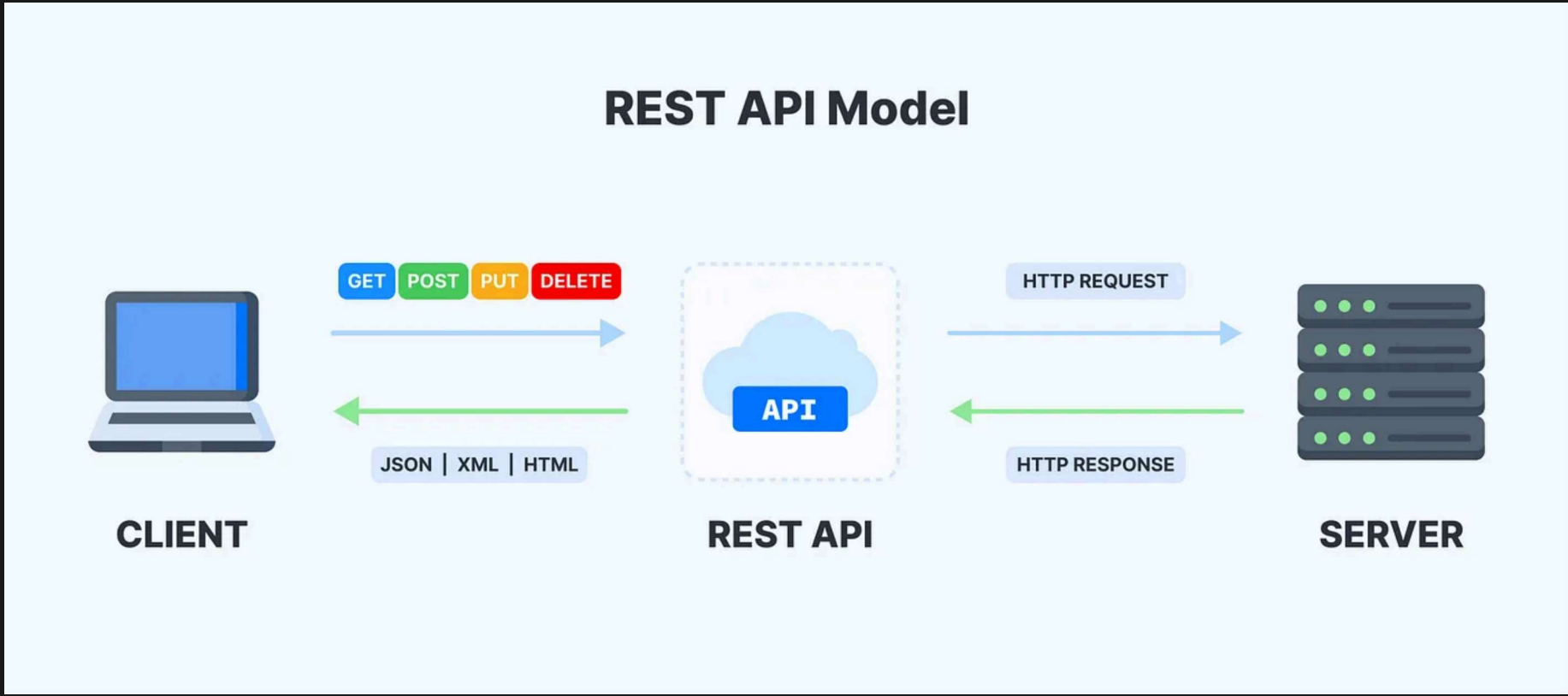
REACT



PART 4: REST API ACCESS

👉 RESTFUL APIS AGAIN 👈



VARIOUS WAYS IN JS / TS... 🧙

- fetch()
- Axios 👈 *find a comparison of them here*

AXIOS

```
npm install axios
```

AXIOS IMPORTS ✅

```
import React, { useEffect, useState } from 'react';
import axios from 'axios';
```

FETCHING DATA WITH REACT

1. prepare an async fetch function

remember: `axios.get` returns a Promise!

```
const [data, setData] = useState(null);
async function fetchData() {
  await axios.get('https://jsonplaceholder.typicode.com/posts')
          .then(response => {
setData(response.data);
          })
          .catch(error => {
console.error('Error fetching data:', error);
          });
};
```

FETCHING DATA WITH REACT

2. apply the useEffect hook 🎣

see Day 3

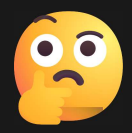
```
useEffect(() => {
  return fetchData
}, []);
```

FETCHING DATA WITH REACT

3. OR do that together 👭

```
const APIComponent = () => {
  const [data, setData] = useState([]);

  useEffect(() => {
    axios.get('https://jsonplaceholder.typicode.com/posts')
      .then(response => {
        setData(response.data);
      })
      .catch(error => {
        console.error('Error fetching data:', error);
      });
  }, []);
  //...
};
```



- `useEffect()` hook takes two arguments
  - setup function to be called when component mounts
  - dependencies array (in case of mount: `[]`)
- `axios.get()` returns a Promise!
- `async` / `await`: useful when fetching data

JSX RETURN

```
  return (
    <div>
      <h2>API Data</h2>
      <ul>
        {data.map(item => (
          <li key={item.id}>{item.title}</li>
        ))}
      </ul>
    </div>
  );
};

export default APIComponent;
```

HANDS ON 💪

1. Pokemon API ⭐
2. Weather API ⭐ ⭐
3. What about POST requests? Build a minimal example ⭐
4. Convert your results to React TS 🤘

CONNECTING TO A LOCALHOST

- Got an endpoint running at http://localhost:8080/api/test
- JSON is returned:

```
{
  "message": "Hello World!",
  "status": 42
}
```

```
useEffect(() => {
  const fetchData = async () => {
    const response = await axios.get(`http://localhost:8080/api/te
    setData(response.data);
    console.log(response.data)
  };
  fetchData();

}, []);
```

the returned Promise is ignored here! 😎

BEEN THERE, DONE THAT 🤭

BUT THERE IS ONE LITTLE THING 👆

CORS

Cross-Origin Resource Sharing

CORS in 100 Seconds

SPRING REST CONTROLLER

there is likely an annotation... 🤰

```
1 @RestController
2 @CrossOrigin(origins = "*", allowedHeaders = "*")
3 public class TestController {
4  // ...
5 }
```

LOCAL REST APIS EXERCISES

1. Frontend: HelloWorld ⭐
2. Frontend: Guestbook API ⭐ ⭐ (including POST requests)
3. Frontend: Horoscope API ⭐ ⭐ (including graphical display)
4. Backend: To-Do List ⭐ ⭐ (store items in a List)
5. Backend: To-Do List ⭐ ⭐ ⭐ (store items in a database), more CORS here?
6. Fullstack: Bookstore application ⭐ ⭐ ⭐